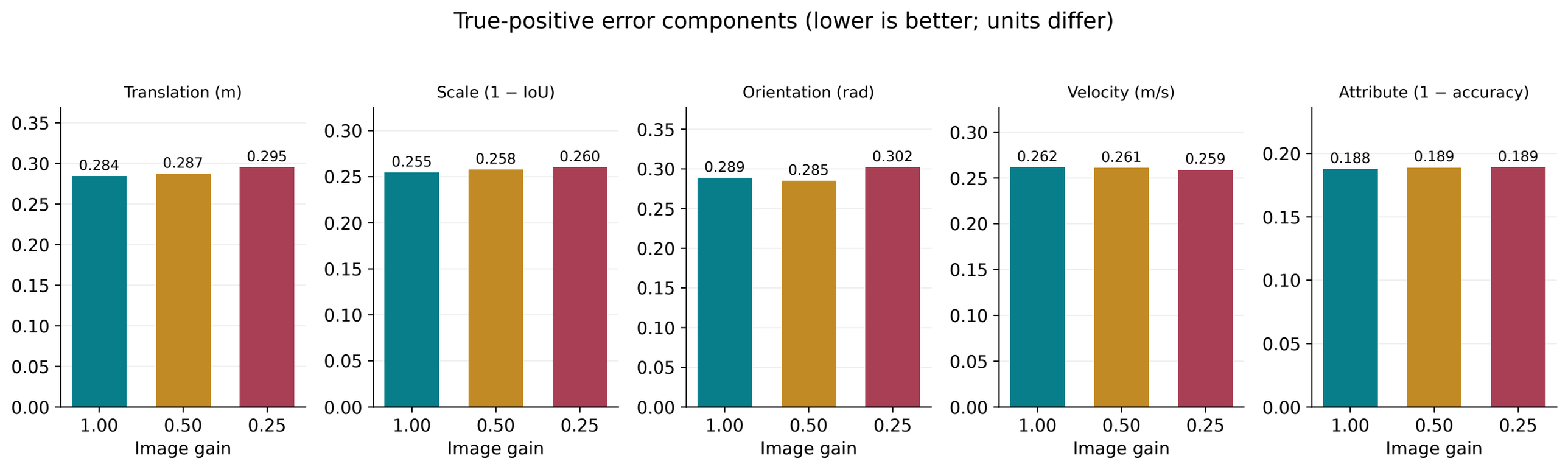
True-positive error components (lower is better; units differ)
Translation (m)
0.284
0.287
0.295
Scale (1 − IoU)
0.255
0.258
0.260
Orientation (rad)
0.289
0.285
0.302
Velocity (m/s)
0.262
0.261
0.259
Attribute (1 − accuracy)
0.188
0.189
0.189
1.00
0.50
0.25
Image gain
600 frozen frames / 150 validation scenes · same pretrained fusion checkpoint · synthetic image darkening